

TARUN REDDY NERELLA

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PROFESSIONAL SUMMARY

AI Data Engineer with experience developing Python, SQL and PySpark pipelines across Databricks, Azure, AWS and Snowflake for batch and streaming workloads exceeding 2 billion records. Combines lakehouse engineering, data quality and observability with applied AI development using RAG, LangGraph, embeddings and REST APIs.

EDUCATION

Master of Science, Computer Science | University of Colorado Denver | GPA: 3.8 | May 2024

CERTIFICATIONS AND PUBLICATION

[Databricks Platform Administrator](#) | [Databricks Certified Generative AI Engineer Associate](#) | [Databricks Certified Data Engineer Professional](#) | [Databricks Certified Data Engineer Associate](#) | [Microsoft Certified Azure Data Engineer Associate](#)

Publication: [A Hybrid Method for Hazardous Asteroid Prediction using XGBoost and Feature Selection](#) | IRJET, Sep 2021

PROFESSIONAL EXPERIENCE

Data Engineer | Hexaware Technologies | Jersey City, NJ | Jul 2024 to Present

- Developed Synapse to Databricks migration using medallion architecture, Delta Live Tables, Lakeflow and Delta MERGE.
- Built Scala, Python and SQL pipelines for 2B+ records with Databricks, ADF, Auto Loader and Structured Streaming.
- Built metadata driven file observability for arrivals, counts, schema changes, SLA status and reprocessing.
- Improved Spark runtime 30 to 40 percent through partition, workflow, dependency and execution plan optimization.
- Used Agent Bricks and coding agents to debug notebooks and validate releases in Azure DevOps and GitHub Actions.

Graduate Student Assistant | CEDC, University of Colorado Denver | Denver, CO | Jan 2023 to May 2024

- Contributed to ML research through Python experiments, reproducible notebooks and model validation.
- Mentored 20+ students in Python, debugging, data structures and code quality through technical reviews.

Data Engineer | 9th Networks Inc. | Hyderabad, India | Aug 2020 to Aug 2022

- Built S3 and Glue PySpark pipelines for schema validation and transformation with direct Snowflake loading.
- Developed MSK and Kafka streaming pipelines, Snowflake models and Python data quality and reconciliation checks.
- Built Plotly and Dash monitoring for arrivals, failures and latency with Docker CI/CD, CloudWatch and IAM.

Full-Stack Developer | IIIT Hyderabad | Hyderabad, India | Jun 2020 to Aug 2020

- Built and optimized a React and Node.js application with REST APIs, backend services and D3.js visualizations.

PROJECTS

Data Contract Observability System | [GitHub](#) | Azure Databricks, Delta Lake, Unity Catalog, Lakeflow, dbt

- Built data contract observability for schema drift, freshness, lineage and SLA failures using dbt tests.

HomeOps AI Native Data Pipeline | [GitHub](#) | GCP, Cloud Run, PostgreSQL, pgvector, LangGraph, MCP

- Built metadata driven ingestion that detects devices, infers schemas, validates events and creates reusable adapters.

ATRIA Human Agent CRM Workflow Platform | [GitHub](#) | Next.js, NestJS, PostgreSQL, Redis, Ollama, BYOK

- Built CRM workflows for tasks, approvals and follow ups with model routing, audit trails and human review.

TECHNICAL SKILLS

Languages: Python, SQL, Scala, JavaScript

Data Engineering: Databricks, PySpark, Spark, Delta Lake, Delta Live Tables, Unity Catalog, Lakeflow, Snowflake, dbt, ETL/ELT

Cloud and Streaming: Azure Data Factory, Synapse, ADLS, AWS S3, Glue, Amazon MSK, Kafka, GCP

AI and Application Development: RAG, LangGraph, embeddings, Databricks Agent Bricks, LLM workflows, REST APIs